

Master Plan of Action (POA)

Architecture, Workflow Engine, Cloud Integrations & AWS Aurora RDS Infrastructure

Status: Approved Execution

Database: AWS Aurora (PostgreSQL)

Date: September 2026

SEQUENTIAL EXECUTION HIERARCHY

Phase 1: Workflows & Canvas Node Boxes Creation (Triggers, Conditions, Actions) | ▼ Phase 2: WhatsApp Cloud API Integration (Templates, Interactive, Chat) | ▼ Phase 3: Facebook Authentication & Lead Generation (OAuth, Pages, Leadgen, CAPI) | ▼ Phase 4: Website Leads Ingestion (Webhooks, Form Handlers, Queue, Attribution) | ▼ Phase 5: Database & Server Infrastructure (AWS Aurora RDS, Multi-Tenancy, Queues) | ▼ Phase 6: Final Corrections, Edge-Case Hardening & End-to-End Testing (QA & Verification)

PHASE 1 Workflows & Canvas Node Boxes Creation (First Priority)

1.1 Visual Canvas Architecture (@xyflow/react)

- **Canvas Engine:** High-performance drag-and-drop workflow canvas with grid snapping, 100px magnetic connection snapping radius, smooth pan/zoom, and minimap.
- **Connection Lines:** Custom step paths with guaranteed pass-through pointer events, preventing canvas interaction blocking.

1.2 Node Box Categories & Design Specifications (Rounded 2xl Cards)

⚡ Trigger Nodes (#F59E0B)

Incoming Lead, Stage / Status Change, Call Completed (duration/disposition), WhatsApp Inbound Keyword, Payment Received, Webhook Inbound.

⚙️ Condition Nodes (#50289B)

Check If Lead / WhatsApp Message, Event Condition / If Else. Dual right handles (green True, red False). Embedded condition filter chips with date-time range selector.

💬 WhatsApp Actions (#008952)

Send WhatsApp Template (dynamic params {{lead.name}}), Interactive CTA Buttons & Phone, Interactive Multi-row Lists, Non-template text/media.

👤 Lead Actions (#EF0024)

Update Lead Assignee (round-robin / pool), Update Lead Fields, Add in List(s), Remove from List(s), Update Lead Status, Update Lead Rating.

🕒 Time Delay (#FB4B60)

Set Delay relative to previous step or date fields (Minutes, Hours, Days, Weeks) with custom variable picker.

⚙️ API & Integrations (#3A2088)

Call API (using stored templates), Meta Conversions API (CAPI offline signal), Add Task, Send Push Notification to Team Member.

1.3 Configuration Drawers & Modals

- **Standardized 580px Drawer:** Clean drawer width across all actions and conditions with top-left circular close (X) button.
- **Slide-Over Condition Picker (AddConditionModal):** Docks to the right edge with real-time search, quick triggers (🔥, 💬, 🕒), and telemetry categories.
- **Live Date & Time Selector:** Dynamic expansion upon date selection: from: 0__:00 h to: 23__:59 h.

PHASE 2 WhatsApp Cloud API Integration

- **Account Provisioning & Credentials:** Multi-tenant isolation for WABA ID, Phone Number ID, and Permanent System Access Tokens.
- **Meta Webhook Subscriptions:** Real-time webhook handlers for messages, message_deliveries, and template_status_update.
- **Template Synchronization (whatsapp_templates):** Auto-syncing Meta-approved templates with dynamic field mapping (Header, Body, CTA & Quick Reply Buttons).
- **Omnichannel Chat (messages):** Two-way conversational messaging directly inside the lead's live timeline with 24h window tracking.

PHASE 3 Facebook Authentication & Lead Generation

- **Facebook OAuth 2.0:** Permissions: pages_show_list, pages_read_engagement, leads_retrieval, pages_manage_ads with automatic exchange for Long-Lived tokens.
- **Connected Pages Table (meta_connected_pages):** Multi-tenant bindings for Facebook Pages & Instagram accounts.
- **Real-Time Lead Ads Webhook:** Webhook listener at /api/integrations/meta/webhook validating SHA-256 HMAC signatures, fetching full leadgen data, deduplicating, and triggering automations.
- **Meta Conversions API (CAPI):** Server-side offline event queue (conversion_events) sending Lead, Schedule, and Purchase events with SHA-256 hashed email/phone for high EMQ.

PHASE 4 Website Leads Ingestion & Attribution

- **Dedicated Webhooks (webhooks):** Secret-verified endpoints compatible with Elementor, WordPress, Contact Form 7, Webflow, Wix, Typeform, Make, and Zapier.
- **JavaScript Tracking Script (pixbe-tracker.js):** Embedded client script capturing UTM parameters (utm_source, utm_medium, utm_campaign), Google Click ID (gclid), Meta Click ID (fbclid), referrer, and landing page.
- **High-Throughput Concurrency Queue (leadIngestionQueue):** Asynchronous queue worker pipeline preventing server stalls during traffic surges, handling deduplication and instant round-robin assignment.

PHASE 5 Database & Server Infrastructure (AWS Aurora RDS)

Complete multi-tenant architecture with 27 relational PostgreSQL tables strictly partitioned by tenant_id:

1. client_tenants	2. company_info	3. agents	4. assignees
5. permission_templates	6. leads	7. pipeline_stages	8. field_settings
9. activity_logs	10. follow_ups	11. campaigns	12. calls
13. reports_call_logs	14. telephone_call_settings	15. workflows	16. actions
17. tasks	18. whatsapp_templates	19. messages	20. templates
21. meta_connected_pages	22. webhooks	23. conversion_events	24. integrations_config
25. payment_transactions	26. tenant_subscriptions	27. documents_quotes	28. ai_sessions

- **Multi-Tenant Failover Service (multiTenantDb.ts):** Dual-driver architecture supporting live AWS Aurora RDS with automatic local store failover.
- **Background Workers:** Meta token renewal worker, call analytics aggregator, and lead queue processors.

PHASE 6 Final Corrections, Edge-Case Hardening & Testing

- **UI/UX Consistency:** Verify color codes (#50289B purple, #EF0024 red, #008952 green), handle ring alignment, and zero clipping across screen resolutions.
- **End-to-End Flow Tests:** WhatsApp template dispatch, interactive button callbacks, Lead Ads webhook verification, CAPI offline matching, and Razorpay webhook validation.
- **Security & Multi-Tenant Audit:** Strict verification of `WHERE tenant_id = $1` on all SQL queries to guarantee zero cross-tenant data leakage.
- **Test Commands:** `npx tsc --noEmit` (Zero TypeScript errors) & `npx ts-node scratch/checkTables.ts` (Aurora schema check).